



MANIPAL SCHOOL OF INFORMATION SCIENCES

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(A constituent unit of MAHE, Manipal)

Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), program are as follows

PEO No.	Education Objective
PEO 1	Develop in depth understanding of the key technologies in data engineering, data science and business analytics.
PEO 2	Practice problem analysis and decision-making using machine learning techniques.
PEO 3	Gain practical, hands-on experience with statistics, programming languages and big data tools through coursework and applied research experiences.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement big data analysis strategies based on theoretical principles, ethical considerations, and detailed knowledge of the underlying data.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing data for decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Algorithms and Data Structures for Big Data [BDE 5101]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Programming with Python or C
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
C01	Implement linked lists, stack, queues and binary search trees.	10	P03	3
C02	Implement dictionary, hash tables, graphs, sorting and searching.	14	P03	3
C03	Analyse the performance of algorithms and implement string and text processing techniques.	12	P04	4



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Introduction to Data Structures and Algorithms	C03
L2	Performance analysis of algorithm - Space	C03
L3	Performance analysis of algorithm - Time	C03
L4	Analysis of recursive programs	C03
L5	Solving recurrence equations.	C03
L6	General Solution for a large class of recurrences	C03
L7	Implementation of Lists	C01
L8	Implementation of Lists	C01
L9	Implementation of Lists	C01
L10	Implementation of Stacks	C01
L11	Implementation of Stacks	C01



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L12	Implementation of Queues	C01
L13	Implementation of Queues	C01
L14	Implementation of Trees	C01
L15	Implementation of Trees	C01
L16	Implementation of Trees	C01
L17	Sorting - bubble, selection and insertion	C02
L18	Sorting - Quick Sort	C02
L19	Sorting - Merge Sort	C02
L20	Sorting - Heap Sort	C02
L21	Searching - Linear and binary	C02
MSE	Mid Semester Exam	C01, C02, C03
L22	Dictionary and Hash Tables	C02
L23	Dictionary and Hash Tables	C02
L24	Hash functions	C02



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L25	Graph Terminology	C02
L26	Representation of Graphs	C02
L27	Traversing Graphing	C02
L28	Minimum Spanning Tree	C02
L29	Shortest Path Algorithm	C02
L30	Shortest Path Algorithm	C02
L31	String and Text Processing	C03
L32	String and Text Processing	C03
L33	String and Text Processing	C03
L34	String and Text Processing	C03
L35	Data Streaming Algorithms	C03
L36	Data Streaming Algorithms	C03



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1.5 References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.
3. Data Structures and Algorithms in Python - Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser. John Wiley & Sons.
4. Data Streams: Algorithms and Applications - S. Muthukrishnan. Foundations and Trends in Theoretical Computer Science archive, Volume 1 Issue 2, August 2005, Pages 117 – 236.
5. MOOCS: <https://www.coursera.org/learn/algorithms-on-strings>

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.